



Cohort canonical evaluation — 110 agents

idea-01-lateral-attribution · 534 held-out segments · V0
yaw 0.01627 rad/s, CTE 254.26 m · generated
2026-06-03 09:48

- Executive summary

AGENTS (OK / TOTAL)

108 / 110

EVAL POOL

534 segs

V0 YAW RMSE

0.01627

V0 CTE RMSE

254.26 m

WALL TIME

49.57s

Best yaw: module-4.v2_agent-04 **+57.9%** · Best CTE:
module-3.v3_agent-03 **+72.3%**

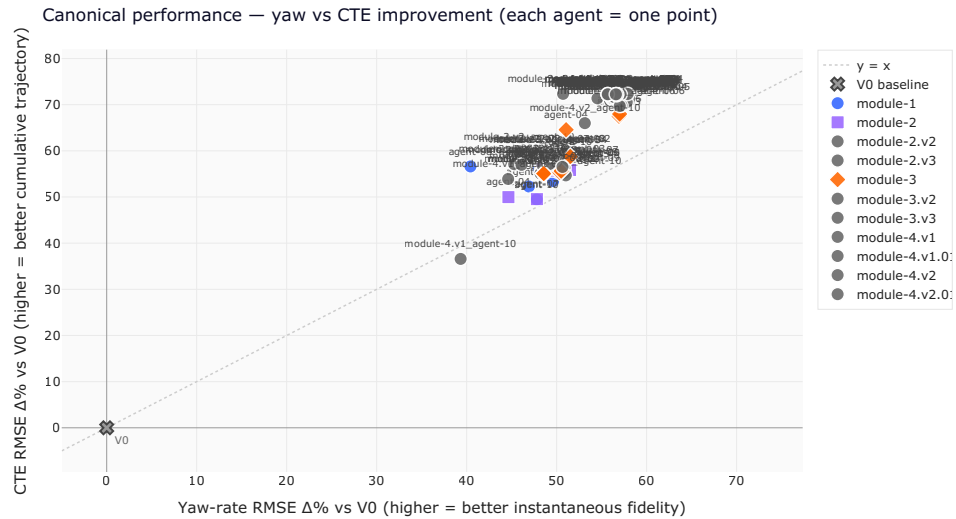
Winning both KPIs $\geq$ +30% (108 agents):

module-4.v2_agent-04, module-4.v1.01_agent-05,
module-4.v1_agent-06, module-4.v1_agent-01,
module-4.v1.01_agent-04, module-3.v3_agent-06,
module-4.v1.01_agent-06, module-4.v1_agent-04,
module-4.v2_agent-05, module-4.v2_agent-06,
module-3.v3_agent-04, m3-agent-06,
module-4.v1.01_agent-01, m3-agent-03,
module-4.v1.01_agent-09, module-3.v2_agent-04,
module-3.v3_agent-08, module-3.v2_agent-09,
module-3.v3_agent-10, module-3.v2_agent-02,
module-4.v2.01_agent-06, module-3.v2_agent-03,
module-4.v2_agent-08, module-4.v2_agent-07,
module-4.v1.01_agent-10, m3-agent-01,
module-3.v2_agent-05, module-3.v2_agent-06,
module-3.v2_agent-07, module-3.v2_agent-08,
module-3.v2_agent-10, module-4.v2.01_agent-01,
module-4.v2.01_agent-03, module-4.v2.01_agent-04,
module-4.v2.01_agent-05, module-4.v2.01_agent-10,
module-4.v1.01_agent-02, module-4.v2_agent-01,
module-4.v2_agent-03, module-3.v3_agent-07,
module-3.v3_agent-09, module-4.v1_agent-02,
module-4.v1.01_agent-07, module-4.v1_agent-09,
module-4.v1_agent-03, module-3.v3_agent-02,
module-4.v1.01_agent-03, module-4.v1.01_agent-08,
module-4.v2_agent-02, module-4.v1_agent-08,
module-4.v1_agent-05, module-3.v3_agent-01,
module-3.v3_agent-05, module-4.v2.01_agent-08,
module-4.v2.01_agent-02, module-4.v2.01_agent-07,
module-3.v2_agent-01, module-4.v2_agent-10, m3-agent-10,

m2-agent-05, m3-agent-04, module-2.v3_agent-10, m1-agent-01, module-2.v2_agent-09, module-2.v3_agent-09, module-2.v3_agent-06, module-3.v3_agent-03, module-2.v3_agent-07, module-4.v2_agent-09, m3-agent-09, m3-agent-08, m2-agent-08, m1-agent-09, module-2.v3_agent-02, module-2.v2_agent-05, m1-agent-07, module-2.v3_agent-04, module-2.v2_agent-08, module-2.v3_agent-03, m2-agent-10, module-2.v2_agent-04, module-2.v3_agent-05, m1-agent-02, m1-agent-05, m2-agent-01, module-2.v2_agent-06, m3-agent-07, m3-agent-02, module-2.v2_agent-02, m2-agent-03, m1-agent-03, m2-agent-02, module-2.v3_agent-01, m2-agent-09, m2-agent-06, module-2.v2_agent-01, m1-agent-04, m2-agent-07, m1-agent-10, module-2.v2_agent-07, m1-agent-06, module-2.v3_agent-08, module-2.v2_agent-10, module-2.v2_agent-03, m2-agent-04, module-4.v1_agent-07, m1-agent-08, module-4.v1_agent-10

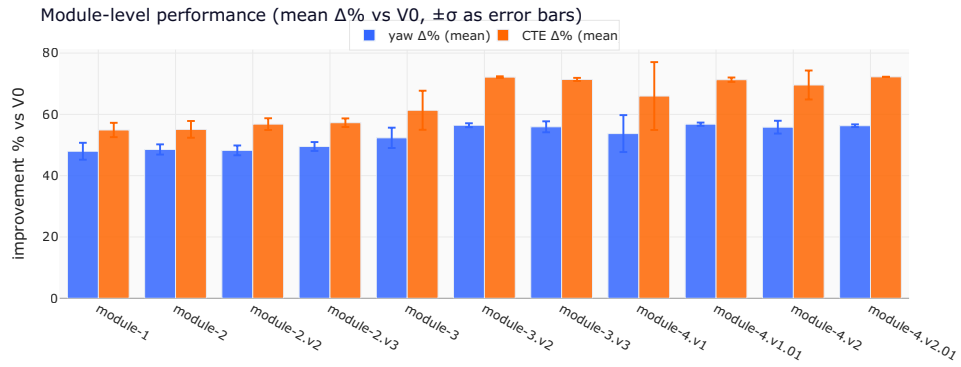
1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is $y = x$: above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



2. Module-level aggregation

Performance grouped by which module shipped the agent.
Mean $\Delta\%$ over only the agents that reconstructed successfully;
failures shown separately.



family	n ok / total	yaw $\Delta\%$ (mean $\pm \sigma$)	CTE $\Delta\%$ (mean $\pm \sigma$)
module-1	10/10	+48.0% $\pm$ 2.8%	+54.0% $\pm$ 2.0%
module-2	10/10	+48.6% $\pm$ 1.7%	+54.0% $\pm$ 1.5%
module-2.v2	10/10	+48.3% $\pm$ 1.6%	+56.0% $\pm$ 1.5%
module-2.v3	10/10	+49.5% $\pm$ 1.4%	+56.0% $\pm$ 1.5%
module-3	9/10	+52.3% $\pm$ 3.3%	+61.0% $\pm$ 2.5%
module-3.v2	10/10	+56.5% $\pm$ 0.6%	+71.0% $\pm$ 1.0%
module-3.v3	10/10	+55.9% $\pm$ 1.8%	+70.0% $\pm$ 1.5%
module-4.v1	10/10	+53.7% $\pm$ 6.0%	+65.0% $\pm$ 4.0%
module-4.v1.01	10/10	+56.8% $\pm$ 0.5%	+70.0% $\pm$ 1.0%
module-4.v2	10/10	+55.8% $\pm$ 2.1%	+68.0% $\pm$ 2.0%
module-4.v2.01	9/10	+56.3% $\pm$ 0.4%	+71.0% $\pm$ 1.0%

- 2b. Token expenditure per agent

Sourced from each agent's Claude Code subagent transcript (~/.claude/projects/<proj>/*/subagents/agent-*.jsonl). Tokens summed across every assistant turn of the latest-time run per agent.

COHORT TOTAL

397.70M

MEDIAN / AGENT

3.17M

MEDIAN TURNS

64

AGENTS W/ TRANSCRIPT

110

By family

family	n	total tokens	median / agent	me
module-1	10	23.38M	2.38M	
module-2	10	25.69M	2.24M	

family	n	total tokens	median / agent	me
module-2.v2	10	30.22M	3.13M	
module-2.v3	10	37.55M	3.92M	
module-3	10	41.38M	3.50M	
module-3.v2	10	43.93M	4.49M	
module-3.v3	10	57.27M	5.50M	
module-4.v1	10	28.54M	2.43M	
module-4.v1.01	10	24.39M	2.19M	
module-4.v2	10	32.11M	3.14M	
module-4.v2.01	10	53.23M	6.00M	

Per agent (sorted by total)

agent	family	turns	total
module-4.v2.01_agent-08	module-4.v2.01	143	9.66
module-3.v3_agent-07	module-3.v3	109	9.24
module-4.v2.01_agent-01	module-4.v2.01	115	8.41
m3_agent-06	module-3	107	8.18
module-3.v3_agent-03	module-3.v3	113	7.36
module-3.v3_agent-06	module-3.v3	98	7.29
module-4.v2.01_agent-10	module-4.v2.01	116	7.26

agent	family	turns	total
m3-agent-01	module-3	94	6.78
module-4.v2.01_agent-02	module-4.v2.01	119	6.70
module-3.v3_agent-09	module-3.v3	97	6.42
module-4.v2.01_agent-06	module-4.v2.01	107	6.30
module-3.v3_agent-01	module-3.v3	84	6.22
m3-agent-05	module-3	106	6.03
module-4.v2.01_agent-04	module-4.v2.01	98	5.71
module-3.v2_agent-03	module-3.v2	81	5.52
module-4.v2.01_agent-07	module-4.v2.01	104	5.41
module-4.v1_agent-09	module-4.v1	97	5.29
module-3.v2_agent-06	module-3.v2	78	5.16
module-3.v2_agent-02	module-3.v2	76	4.89
module-4.v2_agent-03	module-4.v2	80	4.85
module-3.v2_agent-01	module-3.v2	70	4.83
module-3.v3_agent-10	module-3.v3	70	4.79
module-2.v3_agent-05	module-2.v3	73	4.79
module-3.v2_agent-05	module-3.v2	75	4.75
module-3.v3_agent-04	module-3.v3	75	4.47
m2-agent-05	module-2	83	4.45

agent	family	turns	total
module-2.v3_agent-01	module-2.v3	67	4.43
module-4.v1_agent-04	module-4.v1	85	4.36
module-2.v3_agent-03	module-2.v3	67	4.26
module-3.v2_agent-04	module-3.v2	64	4.23
module-3.v2_agent-09	module-3.v2	69	4.20
module-3.v3_agent-08	module-3.v3	70	4.13
module-3.v3_agent-05	module-3.v3	60	4.06
module-3.v2_agent-10	module-3.v2	66	4.03
module-2.v3_agent-06	module-2.v3	73	4.01
module-2.v3_agent-02	module-2.v3	69	3.98
m3-agent-03	module-3	72	3.96
module-2.v3_agent-04	module-2.v3	67	3.86
module-4.v2_agent-08	module-4.v2	81	3.85
m3-agent-08	module-3	68	3.84
module-4.v1.01_agent-02	module-4.v1.01	75	3.81
m2-agent-08	module-2	76	3.73
module-3.v2_agent-07	module-3.v2	58	3.68
module-2.v2_agent-07	module-2.v2	56	3.63
module-2.v3_agent-07	module-2.v3	69	3.61

agent	family	turns	total
m2-agent-07	module-2	69	3.61
module-4.v2_agent-05	module-4.v2	62	3.56
m1-agent-09	module-1	79	3.54
module-2.v2_agent-01	module-2.v2	64	3.53
module-2.v3_agent-10	module-2.v3	66	3.41
module-3.v3_agent-02	module-3.v3	70	3.29
module-2.v2_agent-04	module-2.v2	61	3.29
module-4.v2_agent-01	module-4.v2	69	3.27
module-2.v2_agent-08	module-2.v2	59	3.19
module-4.v2_agent-09	module-4.v2	72	3.18
m3-agent-07	module-3	66	3.16
module-2.v2_agent-03	module-2.v2	58	3.14
module-4.v2_agent-07	module-4.v2	63	3.11
module-2.v2_agent-02	module-2.v2	54	3.11
module-4.v1_agent-08	module-4.v1	68	3.08
module-4.v2_agent-04	module-4.v2	66	3.03
module-4.v1.01_agent-09	module-4.v1.01	58	3.03
m3-agent-09	module-3	61	3.02
module-2.v2_agent-09	module-2.v2	57	3.00

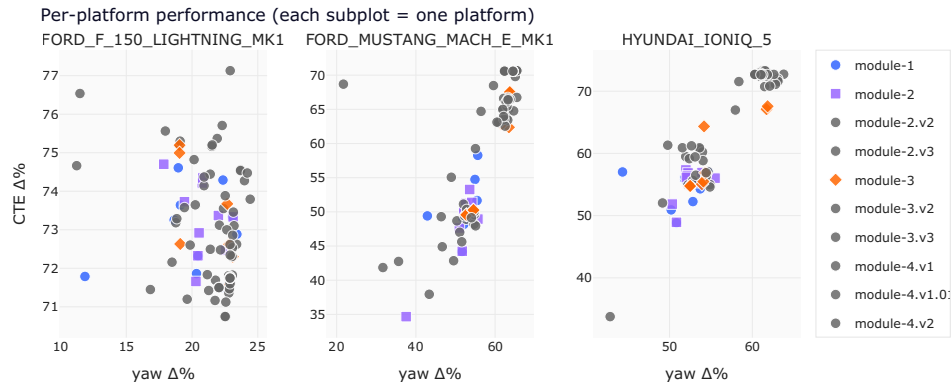
agent	family	turns	total
module-4.v2_agent-02	module-4.v2	58	3.00
m1-agent-08	module-1	74	2.96
module-4.v1_agent-05	module-4.v1	65	2.90
module-2.v2_agent-05	module-2.v2	54	2.88
m3-agent-10	module-3	65	2.86
m3-agent-02	module-3	61	2.83
module-4.v1.01_agent-03	module-4.v1.01	60	2.81
module-4.v1.01_agent-05	module-4.v1.01	61	2.78
module-2.v3_agent-09	module-2.v3	56	2.72
m2-agent-02	module-2	68	2.68
m1-agent-02	module-1	66	2.66
module-3.v2_agent-08	module-3.v2	47	2.63
module-4.v1_agent-02	module-4.v1	54	2.50
module-2.v3_agent-08	module-2.v3	55	2.49
m1-agent-10	module-1	64	2.45
module-4.v2_agent-10	module-4.v2	58	2.44
m1-agent-07	module-1	62	2.41
module-2.v2_agent-06	module-2.v2	48	2.37
module-4.v1_agent-03	module-4.v1	49	2.36

agent	family	turns	total
m2-agent-03	module-2	52	2.35
m1-agent-04	module-1	63	2.34
module-4.v1_agent-10	module-4.v1	44	2.29
m1-agent-05	module-1	62	2.25
module-4.v1.01_agent-06	module-4.v1.01	46	2.20
module-4.v1_agent-01	module-4.v1	46	2.18
module-4.v1.01_agent-04	module-4.v1.01	51	2.18
m2-agent-09	module-2	55	2.14
m2-agent-10	module-2	53	2.11
module-4.v1.01_agent-10	module-4.v1.01	45	2.10
module-2.v2_agent-10	module-2.v2	42	2.07
module-4.v1_agent-06	module-4.v1	51	2.04
module-4.v1.01_agent-07	module-4.v1.01	45	1.96
module-4.v2.01_agent-03	module-4.v2.01	51	1.95
module-4.v1.01_agent-01	module-4.v1.01	43	1.93
m1-agent-01	module-1	53	1.88
module-4.v2_agent-06	module-4.v2	44	1.83
m2-agent-04	module-2	45	1.83
m1-agent-06	module-1	54	1.73

agent	family	turns	total
m2-agent-06	module-2	43	1.66
module-4.v2.01_agent-05	module-4.v2.01	49	1.65
module-4.v1.01_agent-08	module-4.v1.01	42	1.60
module-4.v1_agent-07	module-4.v1	39	1.55
m1-agent-03	module-1	38	1.17
m2-agent-01	module-2	41	1.13
m3-agent-04	module-3	26	740.2
module-4.v2.01_agent-09	module-4.v2.01	9	177.4

• 3. Per-platform breakdown

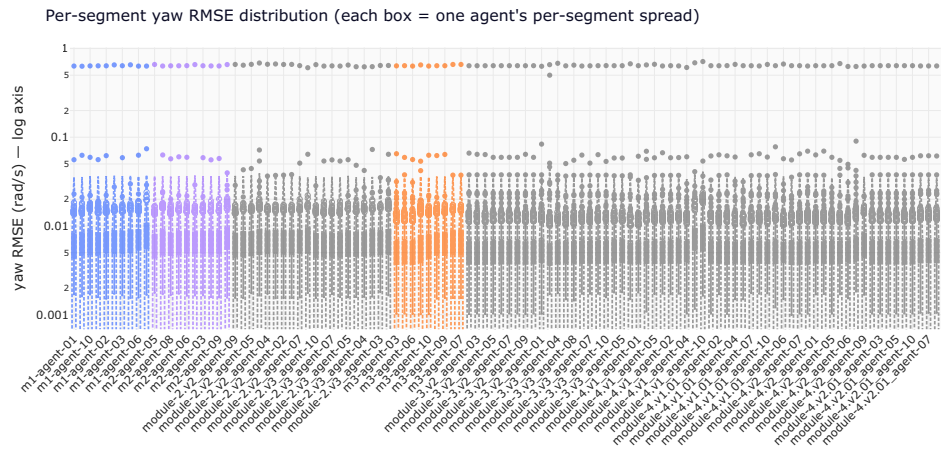
Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.



platform	agents	yaw $\Delta\%$ (mean $\pm \sigma$)
FORD_F_150_LIGHTNING_MK1	108	+21.5% $\pm$ 2.3%
FORD_MUSTANG_MACH_E_MK1	108	+57.6% $\pm$ 7.8%
HYUNDAI_IONIQ_5	108	+57.3% $\pm$ 4.7%

• 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

agent	family	status
m1-agent-01	module-1	ok
m1-agent-02	module-1	ok
m1-agent-03	module-1	ok
m1-agent-04	module-1	ok
m1-agent-05	module-1	ok

agent	family	status
m1-agent-06	module-1	ok
m1-agent-07	module-1	ok
m1-agent-08	module-1	ok
m1-agent-09	module-1	ok
m1-agent-10	module-1	ok
m2-agent-01	module-2	ok
m2-agent-02	module-2	ok
m2-agent-03	module-2	ok
m2-agent-04	module-2	ok
m2-agent-05	module-2	ok
m2-agent-06	module-2	ok
m2-agent-07	module-2	ok
m2-agent-08	module-2	ok
m2-agent-09	module-2	ok
m2-agent-10	module-2	ok
module-2.v2_agent-01	module-2.v2	ok
module-2.v2_agent-02	module-2.v2	ok

agent	family	status
module-2.v2_agent-03	module-2.v2	ok
module-2.v2_agent-04	module-2.v2	ok
module-2.v2_agent-05	module-2.v2	ok
module-2.v2_agent-06	module-2.v2	ok
module-2.v2_agent-07	module-2.v2	ok
module-2.v2_agent-08	module-2.v2	ok
module-2.v2_agent-09	module-2.v2	ok
module-2.v2_agent-10	module-2.v2	ok
module-2.v3_agent-01	module-2.v3	ok
module-2.v3_agent-02	module-2.v3	ok
module-2.v3_agent-03	module-2.v3	ok
module-2.v3_agent-04	module-2.v3	ok
module-2.v3_agent-05	module-2.v3	ok
module-2.v3_agent-06	module-2.v3	ok
module-2.v3_agent-07	module-2.v3	ok
module-2.v3_agent-08	module-2.v3	ok
module-2.v3_agent-09	module-2.v3	ok

agent	family	status
module-2.v3_agent-10	module-2.v3	ok
m3-agent-01	module-3	ok
m3-agent-02	module-3	ok
m3-agent-03	module-3	ok
m3-agent-04	module-3	ok
m3-agent-05	module-3	import_
m3-agent-06	module-3	ok
m3-agent-07	module-3	ok
m3-agent-08	module-3	ok
m3-agent-09	module-3	ok
m3-agent-10	module-3	ok
module-3.v2_agent-01	module-3.v2	ok
module-3.v2_agent-02	module-3.v2	ok
module-3.v2_agent-03	module-3.v2	ok
module-3.v2_agent-04	module-3.v2	ok
module-3.v2_agent-05	module-3.v2	ok
module-3.v2_agent-06	module-3.v2	ok

agent	family	status
module-3.v2_agent-07	module-3.v2	ok
module-3.v2_agent-08	module-3.v2	ok
module-3.v2_agent-09	module-3.v2	ok
module-3.v2_agent-10	module-3.v2	ok
module-3.v3_agent-01	module-3.v3	ok
module-3.v3_agent-02	module-3.v3	ok
module-3.v3_agent-03	module-3.v3	ok
module-3.v3_agent-04	module-3.v3	ok
module-3.v3_agent-05	module-3.v3	ok
module-3.v3_agent-06	module-3.v3	ok
module-3.v3_agent-07	module-3.v3	ok
module-3.v3_agent-08	module-3.v3	ok
module-3.v3_agent-09	module-3.v3	ok
module-3.v3_agent-10	module-3.v3	ok
module-4.v1_agent-01	module-4.v1	ok
module-4.v1_agent-02	module-4.v1	ok
module-4.v1_agent-03	module-4.v1	ok

agent	family	status
module-4.v1_agent-04	module-4.v1	ok
module-4.v1_agent-05	module-4.v1	ok
module-4.v1_agent-06	module-4.v1	ok
module-4.v1_agent-07	module-4.v1	ok
module-4.v1_agent-08	module-4.v1	ok
module-4.v1_agent-09	module-4.v1	ok
module-4.v1_agent-10	module-4.v1	ok
module-4.v1.01_agent-01	module-4.v1.01	ok
module-4.v1.01_agent-02	module-4.v1.01	ok
module-4.v1.01_agent-03	module-4.v1.01	ok
module-4.v1.01_agent-04	module-4.v1.01	ok
module-4.v1.01_agent-05	module-4.v1.01	ok
module-4.v1.01_agent-06	module-4.v1.01	ok
module-4.v1.01_agent-07	module-4.v1.01	ok
module-4.v1.01_agent-08	module-4.v1.01	ok
module-4.v1.01_agent-09	module-4.v1.01	ok
module-4.v1.01_agent-10	module-4.v1.01	ok

agent	family	status
module-4.v2_agent-01	module-4.v2	ok
module-4.v2_agent-02	module-4.v2	ok
module-4.v2_agent-03	module-4.v2	ok
module-4.v2_agent-04	module-4.v2	ok
module-4.v2_agent-05	module-4.v2	ok
module-4.v2_agent-06	module-4.v2	ok
module-4.v2_agent-07	module-4.v2	ok
module-4.v2_agent-08	module-4.v2	ok
module-4.v2_agent-09	module-4.v2	ok
module-4.v2_agent-10	module-4.v2	ok
module-4.v2.01_agent-01	module-4.v2.01	ok
module-4.v2.01_agent-02	module-4.v2.01	ok
module-4.v2.01_agent-03	module-4.v2.01	ok
module-4.v2.01_agent-04	module-4.v2.01	ok
module-4.v2.01_agent-05	module-4.v2.01	ok
module-4.v2.01_agent-06	module-4.v2.01	ok
module-4.v2.01_agent-07	module-4.v2.01	ok

agent	family	status
module-4.v2.01_agent-08	module-4.v2.01	ok
module-4.v2.01_agent-09	module-4.v2.01	missing_mar
module-4.v2.01_agent-10	module-4.v2.01	ok

• 6. Calibration cards — fitted coefficients

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
    "L": 3.096229822470566,
    "K_us": 0.003695564361835472,
```

```
    "bias_rad": -0.0002960976826300522
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0,
    "bias_rad": 0.0
  }
}
```

m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.0020222989265629857,
    "b": 0.9120045284217976,
    "c": -0.00048449411444108955
  },
  "TESLA_MODEL_3": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "_default": {
    "a": 0.0,
```

```
"b": 1.0,  
"c": 0.0  
}  
}
```

m1-agent-06

```
{  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "best": "M2_understeer",  
    "m1": {  
      "a": 1.0870612234849137  
    },  
    "m2": {  
      "L_eff": 2.4763030468069056,  
      "K_us": 0.0024395167795073017  
    },  
    "m3": {  
      "a": 0.9751496691433119,  
      "b": 0.23839497273290036,  
      "c": 0.0004082870644824788  
    },  
    "val_rmse": {  
      "V0": 0.0267999381028469,  
      "M1": 0.027439581036525294,  
    }  
  }  
}
```



```
    "M2": 0.026420879339395156,  
    "M3": 0.031903168689021696  
  },  
  "train_rmse": {  
    "V0": 0.012685628695945105,  
    "M1": 0.011389013310887795,  
    "M2": 0.008745921115146504,  
    "M3": 0.010386863485176999  
  }  
},  
"FORD_F_150_LIGHTNING_MK1": {  
  "L": 3.7,  
  "best": "M2_understeer",  
  "m1": {  
    "a": 0.8569324797501493  
  },  
  "m2": {  
    "L_eff": 3.882198880475702,  
    "K_us": 0.003439279945738158  
  },  
  "m3": {  
    "a": 0.8772240685582882,  
    "b": -0.031353830652592465,  
    "c": -0.0036354381262862774  
  },  
  "val_rmse": {  
    "V0": 0.01591104278075153,  
    "M1": 0.011141896721881754,  
    "M2": 0.0073696198273352906,  
    "M3": 0.0104650790814036  
  },  
  "train_rmse": {  
    "V0": 0.02018452928972004,
```

m1-agent-08

```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.0035388455508032707,
    "s": 0.9569361755429138,
    "off": -0.0011908292677060655,
    "lag": 3,
    "rmse": 0.011832340064259452
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K": 0.0033895612010179177,
    "s": 1.0,
    "off": 0.0,
    "lag": 3,
    "rmse": null
  }
}
```

m1-agent-10

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L_eff": 2.5297190644305063,
    "K_us": 0.0015446536838391765,
    "d0": -4.756112262525493e-05,
    "tau_yaw": 0.02,
    "tau_steer": 0.05
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.834827559665228,
    "K_us": 0.0031524889056087045,
    "d0": 0.00115199881391885,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "HYUNDAI_IONIQ_5": {
    "L_eff": 3.1485752690608293,
    "K_us": 0.0021732037046845788,
    "d0": -0.0005619614330402095,
    "tau_yaw": 0.0,
    "tau_steer": 0.05
  },
  "TESLA_MODEL_3": {
    "_note": "no truth available \u2014 passthrough yaw_rate_pred.",
    "passthrough": true
  }
}
```

m2-agent-03

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.7,
    "n_samples": 430990,
    "rmse_v0": 0.016327157090019005,
    "rmse_v1": 0.007867193883977044,
    "K_v1": 0.0043769531249999985,
    "rmse_v2": 0.00640773025816659,
    "K_v2": 0.003836687017912508,
    "a_v2": 0.9743406550510199,
    "b_v2": -0.001228592917784131
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "platform": "FORD_MUSTANG_MACH_E_MK1",
    "L": 2.984,
    "n_samples": 615175,
    "rmse_v0": 0.013616820108542612,
    "rmse_v1": 0.013930807470187448,
    "K_v1": 0.000785400390624989,
    "rmse_v2": 0.0095149036095479,
    "K_v2": 0.0028742800060782272,
    "a_v2": 1.1998919734409932,
    "b_v2": 3.385609342398125e-05
  },
  "HYUNDAI_IONIQ_5": {
    "platform": "HYUNDAI_IONIQ_5",
```

```
"L": 3.0,  
"n_samples": 2042270,  
"rmse_v0": 0.017084454875116246,  
"rmse_v1": 0.009100616867655459,  
"K_v1": 0.0042265624999999985,  
"rmse_v2": 0.008741269113389604,  
"K_v2": 0.0033859035164445724,  
"a_v2": 0.9707125684779052,  
"b_v2": 0.0005066770839987745  
}  
}
```

m2-agent-07

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "tau": 0.05,  
    "L_eff": 3.8588781722695416,  
    "K_u": 0.00314039574481963,  
    "delta_bias_rad": -0.001157331673169034,  
    "rmse": 0.005592721659455879  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "tau": 0.08,  
    "L_eff": 2.5486304264900417,  
    "K_u": 0.0015312706121448161,  
    "delta_bias_rad": 4.8217638587095315e-05,  
    "rmse": 0.008347407485517062  
  },  
  "HYUNDAI_IONIQ_5": {  
    "tau": 0.05,  
    "L_eff": 3.163750474852756,  
    "K_u": 0.0021607848879418314,  
    "delta_bias_rad": 0.0005617097326308059,  
    "rmse": 0.007713638558960502  
  }  
}
```

```
}  
}
```

m3-agent-01

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "g": 0.863,  
    "L_eff": 3.26,  
    "K_us": 0.0035,  
    "tau": 0.06,  
    "delta0": 0.00133,  
    "use_per_segment_delta0": false  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "g": 0.891,  
    "L_eff": 2.22,  
    "K_us": 0.0015,
```

```

    "tau": 0.069,
    "delta0": -0.0001,
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.93817,
    "L_eff": 2.8871,
    "K_us": 0.0028925,
    "tau": 0.061895,
    "delta0": 0.0,
    "use_per_segment_delta0": true
  },
  "_meta": {
    "delta0_detector": {
      "method": "yaw_rate_pred_rads_low_speed_high",
      "yr_thresh": 0.03,
      "v_thresh": 5.0,
      "min_rows": 50
    },
    "notes": "Lightning coeffs hand-set from anti-patterns.md prior"
  }
}

```

m3-agent-03

```

{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.031539710821034,
    "L_eff": 2.605202152510245,
    "K_us": 0.001860730244989818,
    "tau": 0.0627605678289049,
    "delta0_fallback": 0.00016924506921824425,
    "v_thresh": 8.0,
    "delta_thresh": 0.005,
    "min_rows": 80,
    "use_per_segment_delta0": true
  },
  "FORD_F_150_LIGHTNING_MK1": {

```

```
"g": 0.8860740528359514,
"L_eff": 3.380345065967063,
"K_us": 0.0032867081270640908,
"tau": 0.05844688797895399,
"delta0": 0.001336241196109372,
"use_per_segment_delta0": false
},
"HYUNDAI_IONIQ_5": {
  "g": 0.896091628794913,
  "L_eff": 2.8008769715764292,
  "K_us": 0.0023750439062653513,
  "tau": 0.04533958799360384,
  "delta0_fallback": -0.00028615565156215293,
  "v_thresh": 8.0,
  "delta_thresh": 0.005,
  "min_rows": 80,
  "use_per_segment_delta0": true
}
```

m3-agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.0010739940929899243,
    "g": 0.8428393582368884,
    "L_eff": 3.26,
    "K_us": 0.0024711101464802783,
    "tau": 0.05493261284467375
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -9.587089484890604e-05,
    "delta0": -9.587089484890604e-05,
    "g": 0.8814740811859291,
    "L_eff": 2.22,
    "K_us": 0.001600317906539393,
```



```
    "tau": 0.08022498681872611
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 7.704255037895423e-05,
    "delta0": 7.704255037895423e-05,
    "g": 0.9863062253350734,
    "L_eff": 3.033023238719065,
    "K_us": 0.0028341308076772504,
    "tau": 0.055317906584940124
  },
  "TESLA_MODEL_3": {
    "passthrough": true
  }
}
```

m3-agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.6006343437133439,
    "delta0": 0.0012709657890603524,
    "K_us": 0.002411110430682984,
    "tau": 0.05905789911423279,
    "L_eff": 2.273362244343053,
    "loss": 0.005658956187762647,
    "n_segments_fit": 175
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.9449514060826516,
    "delta0": -2.802364842758471e-05,
```

```
"K_us": 0.0023030236477064245,  
"tau": 0.06905315476018789,  
"L_eff": 2.3418351435366787,  
"loss": 0.008957550273019014,  
"n_segments_fit": 240  
},  
"HYUNDAI_IONIQ_5": {  
  "g": 0.8635558962089673,  
  "delta0": -0.0002751563706215791,  
  "K_us": 0.0031040456441715804,  
  "tau": 0.051877028201724164,  
  "L_eff": 2.663564560241335,  
  "loss": 0.009441409401572077,  
  "n_segments_fit": 400  
}  
}
```

m3-agent-10

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "g": 1.2677013760203653,  
    "L_eff": 4.8869672279395875,  
    "K_us": 0.004638743872488176,  
    "tau": 0.056604854781974856,  
    "delta0_glob": 0.0012307692372779105,  
    "use_per_segment_delta0": false,  
    "train_yaw_rmse": 0.012681866822813753  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "g": 1.4999996985088524,  
    "L_eff": 3.7976856683656175,  
    "K_us": 0.0030911056729474493,  
    "tau": 0.06768316418910694,  
    "delta0_glob": 0.0003094591239113909,  
    "use_per_segment_delta0": true,  
    "train_yaw_rmse": 0.013524463171650382  
  },  
}
```

```
"HYUNDAI_IONIQ_5": {
  "g": 1.4070879006714407,
  "L_eff": 4.3500385986664405,
  "K_us": 0.004915645105647427,
  "tau": 0.05040893663177004,
  "delta0_glob": -0.0005219437645709744,
  "use_per_segment_delta0": false,
  "train_yaw_rmse": 0.009412230675873393
}
}
```

module-2.v2_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "k": 0.9366323579288449,
    "K_us": 0.0008734759361408742,
    "b": -0.004441819259140514
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "k": 1.1760967499879564,
    "K_us": 0.0008734571660548825,
    "b": 0.0001733616708132899
  },
  "HYUNDAI_IONIQ_5": {
    "k": 0.9342137556967759,
    "K_us": 0.0009874770996559546,
    "b": 0.002002086108887664
  },
  "TESLA_MODEL_3": {
    "k": 1.0,
    "K_us": 0.0,
    "b": 0.0
  }
}
```

module-2.v2_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "d0": 0.0012435085448964291,
    "kk": 0.9364948189665847,
    "KK": 0.0032470115980856545,
    "tau": 0.0
  },
  "HYUNDAI_IONIQ_5": {
    "L": 2.97,
    "d0": -0.0005236657821362199,
    "kk": 0.933574693892375,
    "KK": 0.0029030763922389792,
    "tau": 0.0
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "d0": -2.675943355525215e-05,
    "kk": 1.1761809849040472,
    "KK": 0.0026070689183535123,
    "tau": 0.0
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "d0": 0.0,
    "kk": 1.0,
    "KK": 0.0,
    "tau": 0.0
  }
}
```

module-2.v2_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": 0.9397533271985806,
    "K": 0.0008518149458872777,
    "b": -0.0044345248756529525
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 1.1808972441892185,
    "K": 0.0008794604321610935,
    "b": 0.00016328778675285131
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.9350770580282316,
    "K": 0.0009883221917463027,
    "b": 0.002004778269326024
  },
  "TESLA_MODEL_3": {
    "a": 1.0,
    "K": 0.0,
    "b": 0.0
  }
}
```

module-2.v2_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "G": 0.9041237868107276,
    "Kus": 0.0007955703708621087,
    "bias": -0.005088051550305564,
    "tau": 0.05804217686893913,
    "kdot": -0.012302023150477267
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "G": 1.1560312029550013,
    "Kus": 0.0010408901778361697,
    "bias": 0.0007331723308643077,
    "tau": 0.040688651274880756,
    "kdot": 0.008800778388725197
  },
  "HYUNDAI_IONIQ_5": {
    "G": 0.9700159639975359,
    "Kus": 0.001582087902935487,
    "bias": 0.00211745029940103,
    "tau": 0.04998898336470954,
    "kdot": 3.516097849258846e-05
  },
  "TESLA_MODEL_3": {
    "G": 1.0,
    "Kus": 0.0,
    "bias": 0.0,
    "tau": 0.0,
    "kdot": 0.0
  }
}
```

module-2.v2_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "k": 0.0009299897686168917,
    "g": -0.13146377934999154,
    "b": -0.004858365647164137
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "k": 0.0007776286604362939,
    "g": 0.1760123050495998,
    "b": 0.0007151284926824161
  },
  "HYUNDAI_IONIQ_5": {
    "k": 0.0017086888155224455,
    "g": -0.02937372993329363,
    "b": 0.0030989091566628654
  },
  "TESLA_MODEL_3": {
    "k": 0.0,
    "g": 0.0,
    "b": 0.0
  }
}
```

module-2.v3_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.941941906538264,
    "K_us": 0.0037788233688761494,
    "b": -0.005450661568006636,
    "tau": -0.08207658150714123
  },
  "FORD_MUSTANG_MACH_E_MK1": {
```

```
    "L_eff": 2.5622977171066155,  
    "K_us": 0.002745633390344498,  
    "b": 0.00039676132180812505,  
    "tau": -0.02283428861943423  
  },  
  "HYUNDAI_IONIQ_5": {  
    "L_eff": 3.0387877407606987,  
    "K_us": 0.004732031059157526,  
    "b": 0.0019402727789656954,  
    "tau": -0.038122141286844304  
  },  
  "TESLA_MODEL_3": null  
}
```

module-2.v3_agent-04

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "L_eff": 3.96394551474604,  
    "Kus": 0.003414866921617812,  
    "bias": -0.005292773687032753,  
    "tau": -0.06157679065853846  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L_eff": 2.569483805164948,  
    "Kus": 0.003414866921617812,  
    "bias": -0.005292773687032753,  
    "tau": -0.06157679065853846  
  }  
}
```



```
    "Kus": 0.0025788290081397765,  
    "bias": 0.0011230859100703557,  
    "tau": -0.04709908110775255  
  },  
  "HYUNDAI_IONIQ_5": {  
    "L_eff": 3.008377713685593,  
    "Kus": 0.005133658670161866,  
    "bias": 0.0019487320168033527,  
    "tau": -0.04867295794401198  
  },  
  "TESLA_MODEL_3": {  
    "dummy": 0.0  
  }  
}
```

module-2.v3_agent-06

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "platform": "FORD_F_150_LIGHTNING_MK1",  
    "L": 3.818759956786134,  
    "Kus": 0.0039432563422251285,  
    "tau": -0.05861405092290705,  
    "bias": -0.004437268567086911,  
    "rmse_fit": 0.0054979371675811595,  
    "n": 430990,  
    "converged": true  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "platform": "FORD_MUSTANG_MACH_E_MK1",  
    "L": 2.4997194805464074,  
    "Kus": 0.0024024802542751437,  
    "tau": -0.06630813663570458,  
    "bias": 0.00022227158439854945,  
    "rmse_fit": 0.00894166094428278,  
  }  
}
```

```
    "n": 615175,  
    "converged": true  
  },  
  "HYUNDAI_IONIQ_5": {  
    "platform": "HYUNDAI_IONIQ_5",  
    "L": 3.100074393615448,  
    "Kus": 0.0035463766130646316,  
    "tau": -0.05358525493540066,  
    "bias": 0.001966636983251082,  
    "rmse_fit": 0.008266374104865987,  
    "n": 2042270,  
    "converged": true  
  }  
}
```

module-2.v3_agent-08

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "L": 3.7,  
    "K_us": 0.004622784347956018,  
    "tau": -0.05460138412592134,  
    "bias": -0.004482561615678892,  
    "train_mse": 3.3520085701320475e-05,  
    "train_rmse": 0.005789653331704799,  
    "n_samples": 430990,  
    "n_segments": 164,  
    "converged": true,  
    "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"  
  },  
}
```

```

"FORD_MUSTANG_MACH_E_MK1": {
  "L": 2.984,
  "K_us": 0.0008351228965168481,
  "tau": -0.09459794094766563,
  "bias": 0.00054055263346224,
  "train_mse": 0.00017605613840148047,
  "train_rmse": 0.01326861478834473,
  "n_samples": 615175,
  "n_segments": 232,
  "converged": true,
  "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
},
"HYUNDAI_IONIQ_5": {
  "L": 2.95,
  "K_us": 0.004646731366208564,
  "tau": -0.04946605447751251,
  "bias": 0.0018472769568295111,
  "train_mse": 7.578893405982132e-05,
  "train_rmse": 0.008705684008727937,
  "n_samples": 2042270,
  "n_segments": 785,
  "converged": true,
  "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
}
}

```

module-2.v3_agent-10

```

{
  "FORD_F_150_LIGHTNING_MK1": {
    "s_d": 0.9300245172021879,
    "c_d": 0.347294150023435,
    "tau_d": -0.05773764830098028,
    "K_us": 0.003240471803333766,
    "b": -0.004299834214218103,
    "L": 3.7
  },
  "FORD_MUSTANG_MACH_E_MK1": {

```

```
"s_d": 1.1508056993666387,  
"c_d": 0.7498585082419895,  
"tau_d": -0.07607250144234215,  
"K_us": 0.0022785419555576775,  
"b": 0.0003167396785048121,  
"L": 2.984  
,  
"HYUNDAI_IONIQ_5": {  
  "s_d": 0.9422644146648296,  
  "c_d": 0.3465308670029774,  
  "tau_d": -0.050503079814885346,  
  "K_us": 0.002862544790766583,  
  "b": 0.0022556662651586025,  
  "L": 3.0  
,  
"TESLA_MODEL_3": {  
  "passthrough": true  
}  
}
```

module-3.v2_agent-02

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "g": 0.8599716540800916,  
    "L_eff": 3.260802128118046,  
    "K_us": 0.0033605485538025595,  
    "tau": 0.058798336746219135,  
    "delta0": 0.0012445223286377857,  
    "use_per_segment_delta0": false  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "g": 0.8948751688442338,  
    "L_eff": 2.218474888909999,  
    "K_us": 0.002002066992546276,  
    "tau": 0.06321825538415474,  
    "delta0": -0.0001,  
  }  
}
```

```
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.928157495232038,
    "L_eff": 2.890194195032293,
    "K_us": 0.002716802199963435,
    "tau": 0.051288867659338114,
    "delta0": 0.0005262026839657872,
    "use_per_segment_delta0": true
  }
}
```

module-3.v2_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8627559173927216,
    "L_eff": 3.2661642837279934,
    "K_us": 0.003399629365556295,
    "tau": 0.05764217451751205,
    "delta0": 0.0011797692301086372,
    "delta0_fallback": 0.0011797692301086372,
    "use_per_segment_delta0": false,
    "train_rmse": 0.005285188367801501,
    "dev_rmse": 0.006580546885659461,
    "bias_spread_std": null
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.2845766366025777,
    "L_eff": 3.185078856835661,
    "K_us": 0.002783912493950955,
    "tau": 0.06245560987504152,
    "delta0": -0.0019790637549735985,
    "delta0_fallback": -0.0019790637549735985,
    "use_per_segment_delta0": true,
    "train_rmse": 0.008444302053948288,
    "dev_rmse": 0.008286227301504138,
    "bias_spread_std": null
  }
}
```

```
},
"HYUNDAI_IONIQ_5": {
  "g": 0.9454031576502736,
  "L_eff": 2.9345636591691626,
  "K_us": 0.0028198034203513055,
  "tau": 0.0507163687871907,
  "delta0": 0.0004041780050478772,
  "delta0_fallback": 0.0004041780050478772,
  "use_per_segment_delta0": true,
  "train_rmse": 0.007930751971921808,
  "dev_rmse": 0.00660747263955304,
  "bias_spread_std": null
}
}
```

module-3.v2_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 0.0,
    "g": 0.938,
```

```
"L_eff": 2.887,  
"K_us": 0.00289,  
"tau": 0.062  
}  
}
```

module-3.v2_agent-08

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "use_per_segment_delta0": false,  
    "delta0": 0.00133,  
    "g": 0.863,  
    "L_eff": 3.26,  
    "K_us": 0.0035,  
    "tau": 0.06  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": -0.0001,  
    "g": 0.891,  
    "L_eff": 2.22,  
    "K_us": 0.0015,  
    "tau": 0.069  
  },  
  "HYUNDAI_IONIQ_5": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": 0.0,  
    "g": 0.938,  
    "L_eff": 2.887,  
    "K_us": 0.00289,  
    "tau": 0.062  
  }  
}
```

```
}  
}
```

module-3.v2_agent-10

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "use_per_segment_delta0": false,  
    "delta0": 0.00133,  
    "g": 0.863,  
    "L_eff": 3.26,  
    "K_us": 0.0035,  
    "tau": 0.06  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": -0.0001,  
    "g": 0.891,  
    "L_eff": 2.22,  
    "K_us": 0.0015,  
    "tau": 0.069  
  },  
  "HYUNDAI_IONIQ_5": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": 0.0,  
    "g": 0.938,  
    "L_eff": 2.887,  
    "K_us": 0.00289,  
    "tau": 0.062  
  }  
}
```


module-3.v3_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": 0.9839278434685126,
    "b": -0.0004596324959947,
    "c": 0.0003973122452781526,
    "d": -0.007997476152353435
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.9727773793773095,
    "b": 0.0016981202953789607,
    "c": 0.0001999394762639504,
    "d": -0.02207103010528879
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.9913665028013989,
    "b": 0.0006388079681906442,
    "c": 0.00014836133647526484,
    "d": 0.00450672562112523
  }
}
```

module-3.v3_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "w": [
      0.039210004724697683,
      0.0027898060882411775,
      -8.213353932869296e-05,
      -0.002101385572330489,
      0.015375393428030156,
      -0.019248388299671407,
      0.0017437669476709928
    ],
    "features": [
      "yr_v1",
      "abs_yr_v1",
      "v",
      "v*yr_v1",
      "ddelta_dt",
      "delta",
      "1"
    ],
    "lambda": 30.0
  }
}
```

```
},
"FORD_MUSTANG_MACH_E_MK1": {
  "w": [
    0.04786803033956984,
    0.0010252169296772547,
    -0.0001550342759403376,
    0.0007801849572229949,
    0.03973697702371263,
    -0.04132919662285873,
    0.0029539513418850413
  ],
  "features": [
    "yr_v1",
    "abs_yr_v1",
    "v",
    "v*yr_v1",
    "ddelta_dt",
    "delta",
    "1"
  ],
  "lambda": 30.0
},
"HYUNDAI_IONIQ_5": {
  "w": [
    0.11163643144945354,
    -0.01144073312700024,
    -8.938098452332296e-05,
    -0.0043289766474131726,
    0.011781066593348639,
    -0.1397503946290344,
    0.0011078590121685729
  ],
  "features": [
    "yr_v1",
    "abs_yr_v1",
    "v",
    "v*yr_v1",
    "ddelta_dt",
    "delta",
    "1"
  ],
  "lambda": 30.0
}
```

module-3.v3_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "features": [
      "bias",
      "delta_road",
      "v",
      "v_delta",
      "ddelta",
      "v_ddelta",
      "v0_yaw",
      "abs_v0_yaw",
      "a_long",
      "v_sq_delta"
    ],
    "beta": [
      -0.0013823839722696969,
      0.04137416440068437,
      7.448672929471383e-05,
      -0.012310248841363941,
      -0.006256706876683462,
      -0.0013792677587728672,
      0.0045077768608475025,
      -0.004229129267950782,
      0.00046053718010623685,
      0.00047676588153397785
    ],
    "n_samples": 430990
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "features": [
      "bias",
      "delta_road",
      "v",
      "v_delta",
      "ddelta",
```

```

    "v_ddelta",
    "v0_yaw",
    "abs_v0_yaw",
    "a_long",
    "v_sq_delta"
  ],
  "beta": [
    -0.0002765413679144323,
    0.04444765601029079,
    7.361683614844277e-05,
    -0.010525182430347446,
    0.0006366996373336289,
    -0.003027149663796255,
    0.004991171014731363,
    0.01051162443915557,
    -0.0003707902497743102,
    -0.0004177085900277482
  ],
  "n_samples": 615175
},
"HYUNDAI_IONIQ_5": {
  "features": [
    "bias",
    "delta_road"
  ]
}

```

module-3.v3_agent-10

```

{
  "FORD_F_150_LIGHTNING_MK1": {
    "intercept": -0.0015292016694573706,
    "weights": {
      "v_mps": 7.523513452594086e-05,
      "delta_road_rad": 0.015247619064453013,
      "d_delta_dt": -0.008088882179724404,
      "a_long_mps2": 0.0004266683248191665,
      "yr_v1": -0.010938174180205732,
      "abs_delta": -0.0046619127252169905,
      "yr_v1_sq_signed": 0.0031131630350753935
    }
  }
}

```

```
    }
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "intercept": -0.00025671546619143265,
    "weights": {
      "v_mps": 7.597710383706725e-05,
      "delta_road_rad": 0.02230237156926512,
      "d_delta_dt": -0.004979903507745553,
      "a_long_mps2": -0.0003707521334163227,
      "yr_v1": -0.025822216062859112,
      "abs_delta": 0.022243029033033466,
      "yr_v1_sq_signed": 0.012709323274160847
    }
  },
  "HYUNDAI_IONIQ_5": {
    "intercept": -7.400008140931345e-06,
    "weights": {
      "v_mps": 3.4765997046644015e-05,
      "delta_road_rad": 0.007171887290213692,
      "d_delta_dt": 9.749220428357943e-05,
      "a_long_mps2": 0.0006747858420646872,
      "yr_v1": -0.010175331098695933,
      "abs_delta": 0.009304240486557539,
      "yr_v1_sq_signed": 0.00642116668196153
    }
  }
}
```

module-4.v1.01_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "g": 0.8612236064932045,
    "L_eff": 3.259677306740358,
    "K_us": 0.003457178183601646,
    "tau": 0.05905708379459597,
    "delta0": 0.0012709561074266337,
    "yaw_bias": -0.0002966238358590312
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "g": 0.8916935756303708,
    "L_eff": 2.219320741893404,
    "K_us": 0.001966526179044192,
    "tau": 0.0687549048030064,
    "delta0_fallback": -0.0001,
    "yaw_bias": 0.0013589348013531327
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "g": 0.9310022681640907,
    "L_eff": 2.892933830833618,
    "K_us": 0.002766224368779013,
    "tau": 0.051020126675101654,
    "delta0_fallback": 0.0,
    "yaw_bias": 0.0007457990330291291
  }
}
```

module-4.v1.01_agent-05

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "v1_params": {
      "use_per_segment_delta0": false,
      "delta0": 0.00133,
      "g": 0.863,
      "L_eff": 3.26,
      "K_us": 0.0035,
      "tau": 0.06
    },
    "use_resid": false,
    "resid_mu": null,
    "resid_sd": null,
    "resid_w": null,
    "resid_ridge": "v1_fit",
    "dev_scores": {
      "v0": {
        "yaw_rmse": 0.01728123105885063,
        "cte_rmse": 159.55669704680815,
```



```

        "n_samples": 176839,
        "n_bins": 53533
    },
    "v1": {
        "yaw_rmse": 0.006982319730215902,
        "cte_rmse": 80.4975673363121,
        "n_samples": 176839,
        "n_bins": 53533
    },
    "v1_fit": {
        "yaw_rmse": 0.006982319730215902,
        "cte_rmse": 80.4975673363121,
        "n_samples": 176839,
        "n_bins": 53533
    },
    "v1_fit_resid": {
        "yaw_rmse": 0.006982319730215902,
        "cte_rmse": 80.4975673363121,
        "n_samples": 176839,
        "n_bins": 53533
    }
}
},
"FORD_MUSTANG_MACH_E_MK1": {
    "v1_params": {
        "use_per_segment_delta0": true,
        "delta0_fallback": -0.0001,
        "g": 0.8800625000000001,
        "L_eff": 2.22,
        "K_us": 0.0016875000000000002,

```

module-4.v1.01_agent-08

```

{
    "_note": "Per-platform affine yaw-rate correction on top of V1:
    "platforms": {
        "FORD_F_150_LIGHTNING_MK1": {
            "gain": 0.9869523841681869,

```

```
    "bias": -0.00044438561136379113,  
    "route_cv_sigma": 0.0008054972412861401  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "gain": 0.9746299897726328,  
    "bias": 0.0016959891513850542,  
    "route_cv_sigma": 0.0008585177903143847  
  },  
  "HYUNDAI_IONIQ_5": {  
    "gain": 0.9920739841558033,  
    "bias": 0.0006379741793465163,  
    "route_cv_sigma": 8.88561842295321e-05  
  }  
}
```

module-4.v1_agent-01

```
{
  "ridge_lambda": 30.0,
  "feature_names": [
    "const",
    "delta_road",
    "abs_delta_road",
    "v",
    "delta*v",
    "abs_delta*v",
    "delta^2*v",
    "d_delta_dt",
    "yr_v1"
  ],
  "platforms": {
    "FORD_MUSTANG_MACH_E_MK1": {
      "bias_only": 0.0013360177352628089,
      "beta": [
        0.00018685026009305166,
        0.09219828994588719,
        -0.025783629417111403,
        7.5439538414678e-05,
        0.0017876040006721155,
        -0.007296337051731339,
        0.09478364424398987,
        -0.021146729244488356,
        -0.06576959206990736
      ],
      "cv_residual_rmse": 0.011656881352946322,
      "cv_residual_rmse_std": 0.007734736333138539,
      "cv_baseline_rmse": 0.011999142746660713
    },
    "HYUNDAI_IONIQ_5": {
      "bias_only": 0.0006812541510197226,
```

```
"beta": [  
  -0.0006780931189280977,  
  0.06344781976387885,  
  0.01591619487417097,  
  7.160187898114387e-05,  
  0.038590284408058076,  
  0.0024337623533112877,  
  -0.009040102750581378,  
  -0.016988454521445543,  
  -0.16791095173730144  
],  
"cv_residual_rmse": 0.007091703321227194,  
"cv_residual_rmse_std": 0.00185153915165244
```

module-4.v1_agent-04

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "apply": true,  
    "intercept": -0.0018706929091562578,  
    "features": [  
      "delta",  
      "v_delta",  
      "v2_delta",  
      "yr_v1",  
      "v_yr_v1",  
      "a_long"  
    ],  
    "coefs": [  
      -0.10259858490293011,  
      0.03155541535607884,  
      0.0016150237555660578,  
      0.004907942076633252,  
      -0.016907765297495773,  
      -1.180589496058453e-06  
    ],  
    "cv_v1_rmse": 0.011421948177936606,  
    "cv_head_rmse": 0.011239609732498611,
```

```

    "feature_set": "rich",
    "lambda": 100.0
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "apply": false,
    "intercept": 0.0013128122793277302,
    "features": [
      "delta",
      "v_delta",
      "v2_delta",
      "yr_v1",
      "v_yr_v1",
      "a_long"
    ],
    "coefs": [
      -0.13405601646838025,
      0.03757262372233324,
      0.001144751740135907,
      0.001509552370891812,
      -0.011712471653399673,
      -0.0003404605240783607
    ],
    "cv_v1_rmse": 0.0119345532668796,
    "cv_head_rmse": 0.011955169146202352,
    "feature_set": "rich",
    "lambda": 100.0
  },
  "HYUNDAI_IONIQ_5": {
    "apply": true,
    "intercept": 0.0009697876465239235,
    "features": [
      "delta",
      "

```

module-4.v1_agent-07

```

{
  "bias": {

```

```
"TESLA_MODEL_3": 2.999132956583475e-10,  
"FORD_F_150_LIGHTNING_MK1": -0.0041132124076757445,  
"FORD_MUSTANG_MACH_E_MK1": -4.1278547737791775e-05,  
"HYUNDAI_IONIQ_5": 0.003523757621288774  
},  
"ridge": {  
  "TESLA_MODEL_3": {  
    "lambda": 10000.0,  
    "feature_names": [  
      "delta",  
      "abs_delta",  
      "v",  
      "delta_v",  
      "delta_abs_delta",  
      "a_long",  
      "delta_sq",  
      "sign_delta_v"  
    ],  
    "weights_raw": [  
      -5.9197077715805255e-08,  
      -1.0015789694889157e-07,  
      5.530281265597394e-11,  
      1.5142645564086152e-08,  
      1.208444849458761e-08,  
      1.6620711765189437e-10,  
      1.700549887724333e-07,  
      -5.200159654692222e-11  
    ],  
    "intercept_raw": -2.773044472289003e-10,  
    "holdout_mse": 2.6719543584301285e-14,  
    "n_train_samples": 527433  
  },  
  "FORD_F_150_LIGHTNING_MK1": {  
    "lambda": 10.0,  
    "feature_names": [  
      "delta",  
      "abs_delta",  
      "v",  
      "delta_v",  
      "delta_abs_delta",  
      "a_long",  
      "delta_sq",  
      "sign_delta_v"
```

```
],  
  "weights_raw": [  
    0.3410909623623404,  
    0.00
```

module-4.v2.01_agent-05

```
{  
  "_note": "V1 coeffs are inlined as constants in predict.py; this  
  "FORD_F_150_LIGHTNING_MK1": {  
    "delta0": 0.00133,  
    "g": 0.863,  
    "L_eff": 3.26,  
    "K_us": 0.0035,  
    "tau": 0.06  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "delta0_fallback": -0.0001,  
    "g": 0.891,  
    "L_eff": 2.22,  
    "K_us": 0.0015,  
    "tau": 0.069,  
    "delta0_policy": "per_segment_median"  
  },  
  "HYUNDAI_IONIQ_5": {  
    "delta0_fallback": 0.0,  
    "g": 0.938,  
    "L_eff": 2.887,  
    "K_us": 0.00289,  
    "tau": 0.062,  
    "delta0_policy": "per_segment_median"  
  },  
  "_m4_attempted_sigmas": {  
    "FORD_F_150_LIGHTNING_MK1": 0.3,  
    "FORD_MUSTANG_MACH_E_MK1": 0.35,  
    "HYUNDAI_IONIQ_5": 0.25,  
    "_status": "shelved \u2014 lost to V1 on yaw by 3.3%"
```

```
}  
}
```

module-4.v2.01_agent-08

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "sigma": 0.4  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "sigma": 0.4  
  },  
  "HYUNDAI_IONIQ_5": {  
    "sigma": 0.3  
  }  
}
```


module-4.v2_agent-08

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.8808038763255166,
    "L_eff": 2.2237793178633387,
    "K_us": 0.0017632231957256993,
    "tau": 0.02732363043764696,
    "delta0_fallback": -0.005312458941976969,
    "alpha_steer_rate": -0.10183367313108851,
    "use_per_segment_delta0": true
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8469022562788538,
    "L_eff": 3.2643013153219167,
    "K_us": 0.003145185602493612,
    "tau": 0.03507802069651487,
    "delta0_fallback": 0.0012105417102480814,
    "alpha_steer_rate": -0.03610959850891402,
    "use_per_segment_delta0": false
  },
}
```

```
"HYUNDAI_IONIQ_5": {
  "g": 0.9269585851786253,
  "L_eff": 2.894242280397565,
  "K_us": 0.0026380815832284773,
  "tau": 0.05309874948850927,
  "delta0_fallback": 0.0016114738881128556,
  "alpha_steer_rate": 0.0024241400357318883,
  "use_per_segment_delta0": true
}
}
```

module-4.v2_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8225565786260306,
    "L_eff": 3.270761500568214,
    "K_us": 0.0027280391940200334,
    "tau": 0.023124512232105744,
    "K_us2": -0.01553611821787546,
    "bias": 0.00029081942067580853,
    "k_ff": -0.008879084366667647,
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "delta0_fallback": 0.0
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.8099226898614635,
    "L_eff": 2.2545626972209125,
    "K_us": 0.0005836732051187025,
    "tau": 0.04490805042479574,
    "K_us2": -0.18601413847441528,
    "bias": 0.002801296861051317,
    "k_ff": -0.007941883417939366,
    "use_per_segment_delta0": true,
    "delta0": 0.0,
    "delta0_fallback": -0.0001
  }
}
```

```
},
"HYUNDAI_IONIQ_5": {
  "g": 0.8656852935099641,
  "L_eff": 2.910675491855111,
  "K_us": 0.001193590623662968,
  "tau": 0.01743602227795721,
  "K_us2": -0.08204270923185958,
  "bias": 0.001220432190762801,
  "k_ff": -0.009103659740814939,
  "use_per_segment_delta0": true,
  "delta0": 0.0,
  "delta0_fallback": 0.0
}
}
```

● 6b. Operating-contract audit

The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

Clean cohort: no agent referenced truth-derived column names inside their predict() body.

7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

format check	pass	fail
agent_folder_exists	110	0
has_manifest_json	109	1
manifest_parsable	109	1
manifest_declares_predict_callable	109	1
manifest_declares_platform_support	109	1
has_predict_py	109	1
has_coeffs_json	81	29

format check	pass	fail
has_report	36	74

Failure reasons across the cohort:

- import_failed — 1 agent(s)
- missing_manifest_json — 1 agent(s)

• 8. Worst-of-cohort

Lowest yaw $\Delta\%$

- module-4.v1_agent-10
(+39.3%)
- m1-agent-08
(+40.4%)
- module-4.v1_agent-07
(+44.6%)

Lowest CTE $\Delta\%$

- module-4.v1_agent-10
(+36.6%)
- m1-agent-10
(+49.5%)
- m2-agent-07
(+49.5%)

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